Prithvi's Notebook

Home Insert Draw View

Monday, November 17, 2025 9:10 AM

Introduce the problem statement → Agenda (what we will be covering in the presentation) → Background info (introduce the ruleset, the sensor modalities, and the current state of Artemis)

→ Regulations we have to consider (our chosen 5 locations, what regulations we fail with our current sensor suite) → Our MVP (The sensors on the market, our placements, the FOV, total cost, power, milleage, etc.)

→ ODD & self certification (a lot of info will be hidden slides)

# Logos

aUToronto

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# aUToronto

R2Y5 0-0-0 Challenge - Sensor Suite Optimization

By: Prithvi Seran, Amanda Liu, Forest Li, Nevan Kho, Ailing Ji

# For the R2Y5 competition, all teams had to engineer a sensor suite for the car.

# There's a problem...

# These sensor suites end up being impractical!

# Optimize the sensor suite of your car.

# R2Y5 RULESET

Extra rules we have to consider in addition to our problem statement

### Compliance with specific regulations

FMVSS111, FMVSS127, FMVSS208

### Compliance with regulations in 5 states/provinces

California, Michigan, Texas, Arizona, Ontario

### Benchmark analysis with another robotaxi

We have chosen Waymo and Tesla

**Assumption:** We will only be optimizing for regulatory requirements and road legality.

# AGENDA

- Problem statement
- Background information
- Regulations
- MVP
- Self-certifications
- Key learnings

# Background Information

Problem statement | Background information | Regulations | MVP | Self-certifications | Key learnings

# Sensor Modalities

**Cameras**

2D image capturing for recognizing road signs, lane markings, and colors

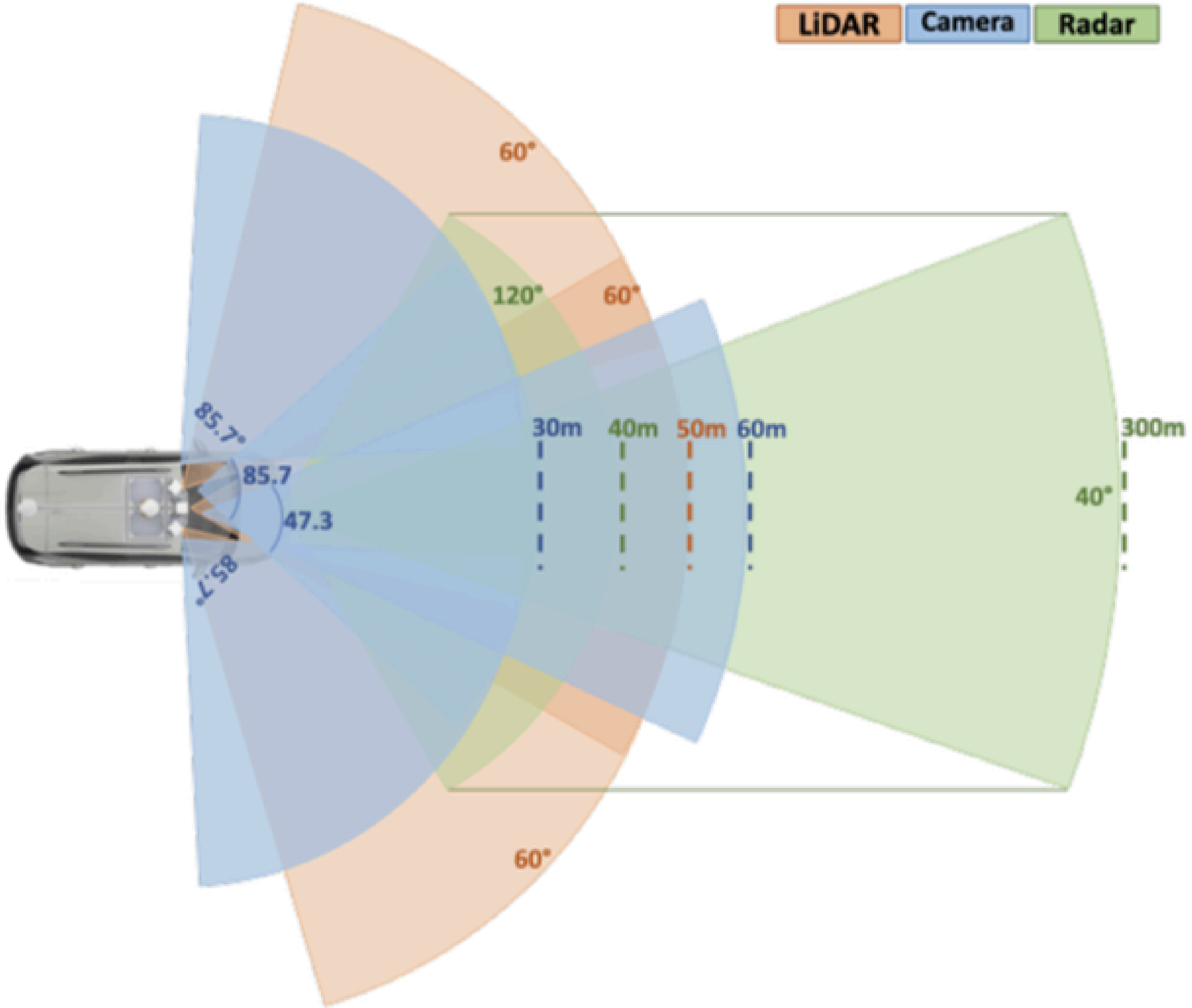
**LiDAR**

Detects obstacles, road features, and pedestrians using 3D point clouds

**Radar**

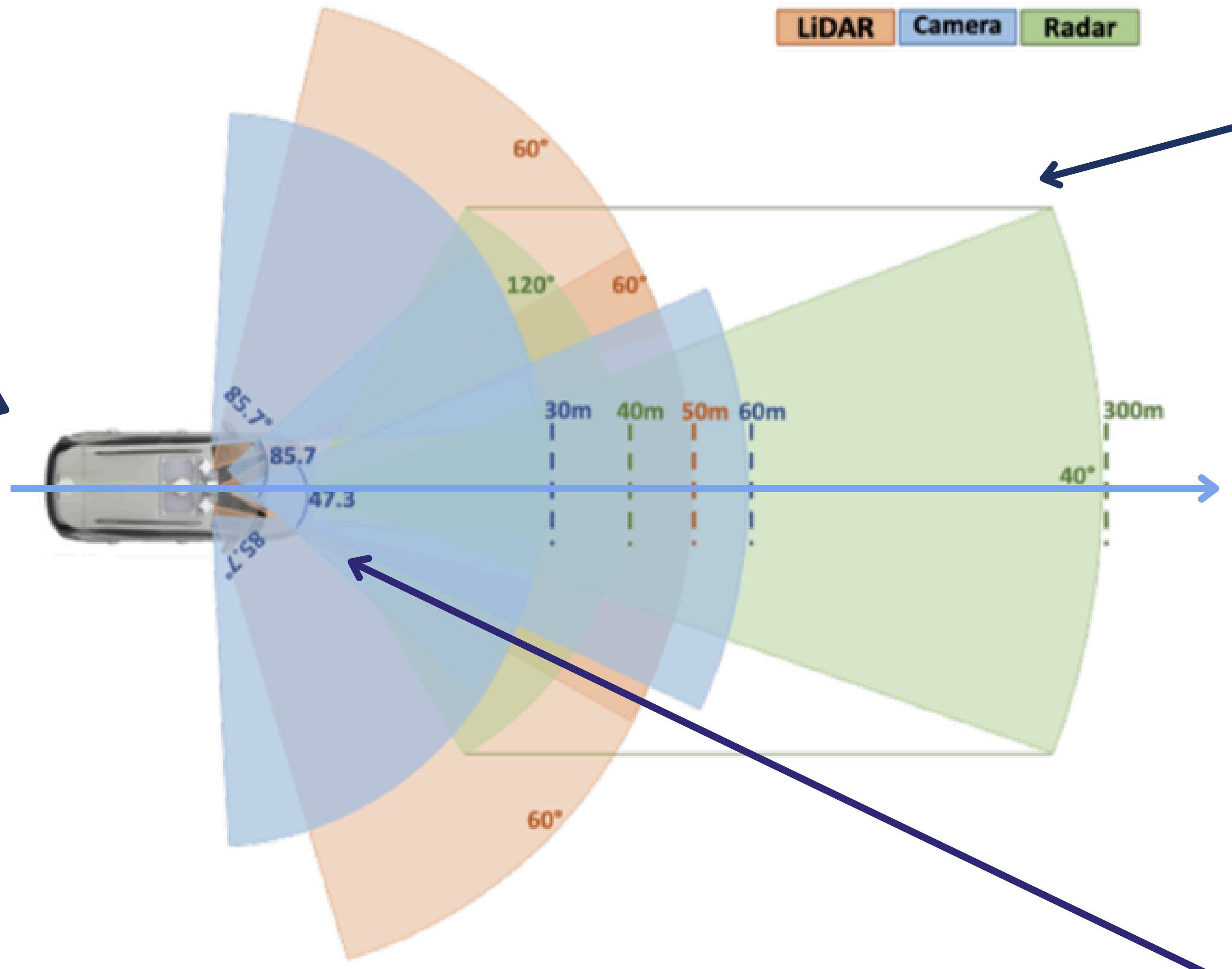
Tracks objects and their speed using doppler effects

# Our R2Y5 Sensor Suite



# A lot of room for improvement...

Completely no coverage on the back of the car




300 meter range is not needed

Lots of overlap

# Regulatory Compliance

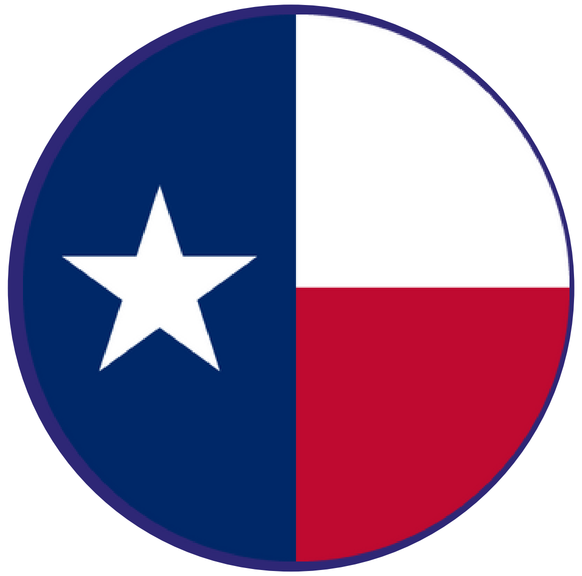
Problem statement → Background information → Regulations → MVP → Self-Certifications → Key learnings

# OUR STATES & PROVINCE

Required States


California, USA


Michigan, USA



Texas, USA

Additional 2

Arizona, USA

Toronto, CAN

# OUR SCENARIO & ASSUMPTION

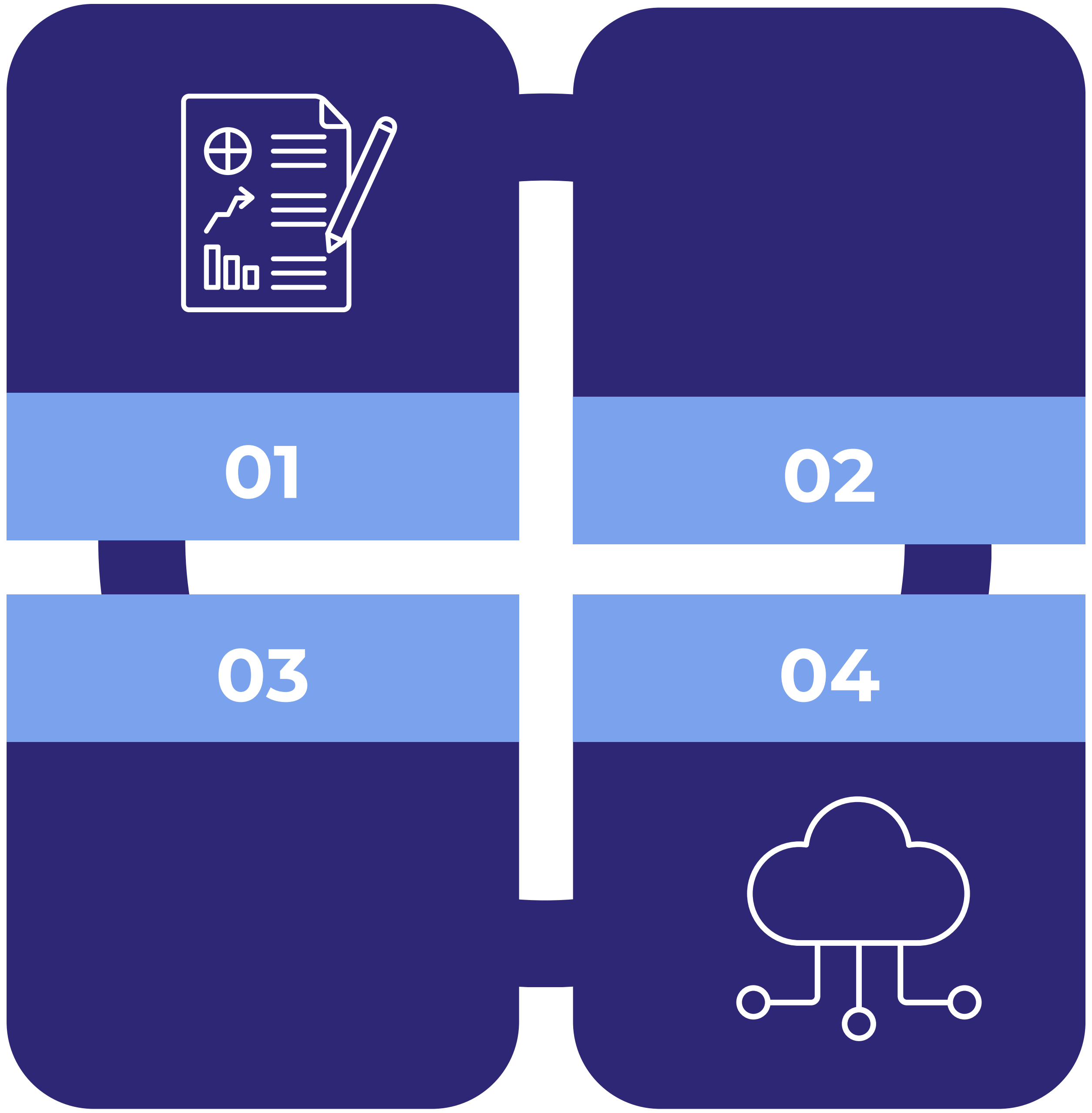
We comply  with the standards

## Context

Our focus for regulatory compliance will be on the **sensors**, and not the autonomous car as a whole.

## Physical Car

We will be assuming that the underlying car (Chevrolet EUV Bolt) already complies with all the regulations.



# MAIN REGULATORY BODIES

# FEDERAL MOTOR VEHICLE SAFETY STANDARDS (FMVSS)

NOTE: Only for the US [SAE AutoDrive R2Y5 Ruleset]

**FMVSS 111**: Regulation for **Rear Visibility Requirements,** including equipment of rearview camera that provides a clear field of view behind the vehicle when in reverse.

**FMVSS 127**: Regulation for **Automatic Emergency Braking** and Pedestrian Automatic Emergency Braking in both daylight and darker conditions at night.

**FMVSS 208**: Regulation for **Occupant Crash Protection** design and performance of safety systems in motor vehicles. Includes seat belts, airbags, crash testing, and child restraint systems.

# DMV REGULATIONS

NOTE: Only for the US [SAE AutoDrive R2Y5 Ruleset]

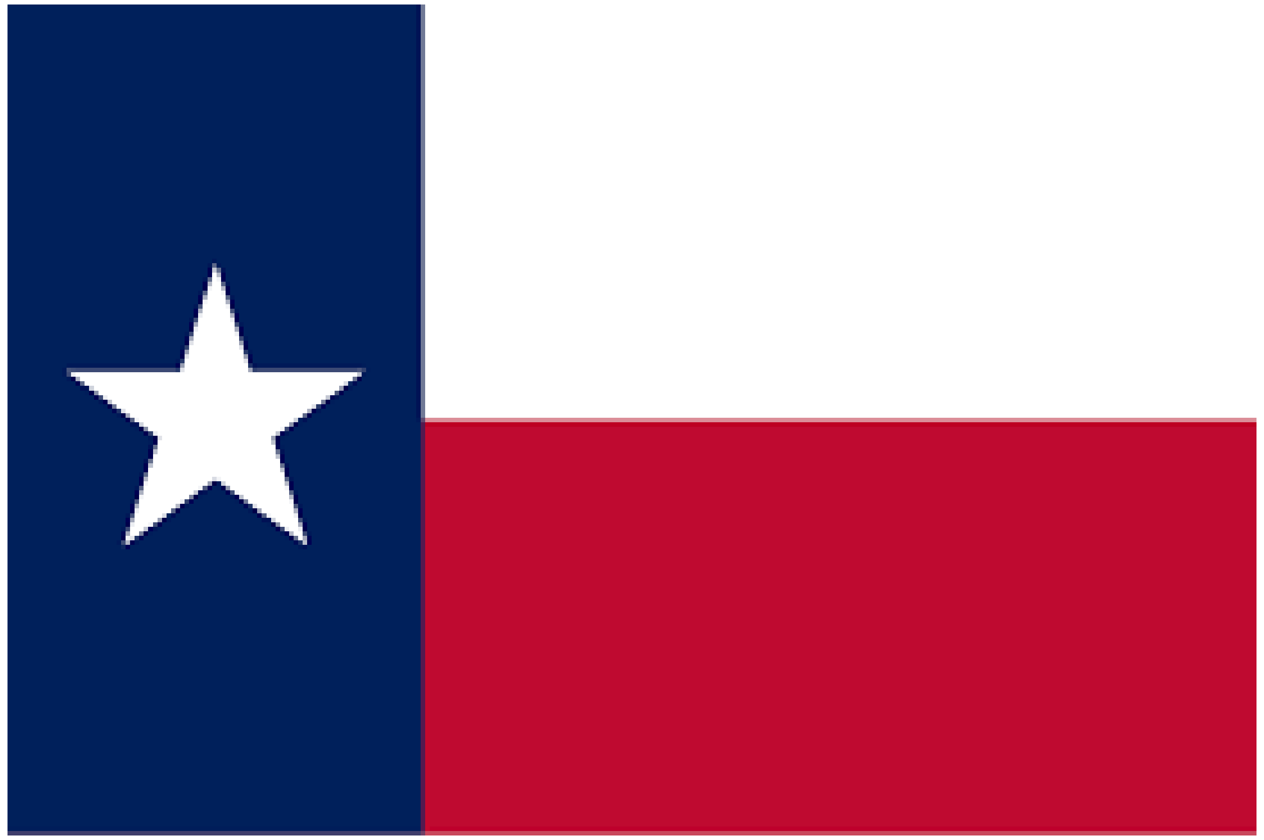

1. Provide an easily accessible method for the driver to engage or disengage the autonomous driving system.
2. Include an in-vehicle indicator that informs the driver when the autonomous driving system is active.
3. Have a system that alerts the driver if the autonomous driving system fails and either allows manual takeover or safely brings the vehicle to a complete stop.
4. Comply with all applicable Federal Motor Vehicle Safety Standards.
5. Include a separate mechanism to record and store sensor data from at least 30 seconds before a collision.

# STATE OF CALIFORNIA

Clearing all DMV requirements is enough for California!

# STATE OF TEXAS

**Alongside clearing the DMV requirements...**



1. Need to create a first responders safety plan (more on our self certification soon!)
2. The DMV must have enhanced authority to oversee autonomous vehicle operations
3. Need to report safety data and incidents [20]

# STATE OF ARIZONA

**Alongside clearing the DMV requirements...**

1. Must submit a plan to ADOT and the Arizona Department of Public Safety
   - explains how law enforcement should interact with your autonomous vehicles during emergencies, traffic stops, and accidents.
2. A self certification acknowledging:
   - Compliance with all federal safety standards.
   - Compliance with all Arizona traffic and motor vehicle safety laws. [23, 24, 25]
   - can achieve a “minimal risk condition” (i.e., safely stopping) in case of system failure.

# STATE OF MICHIGAN

**Alongside clearing the DMV requirements...**

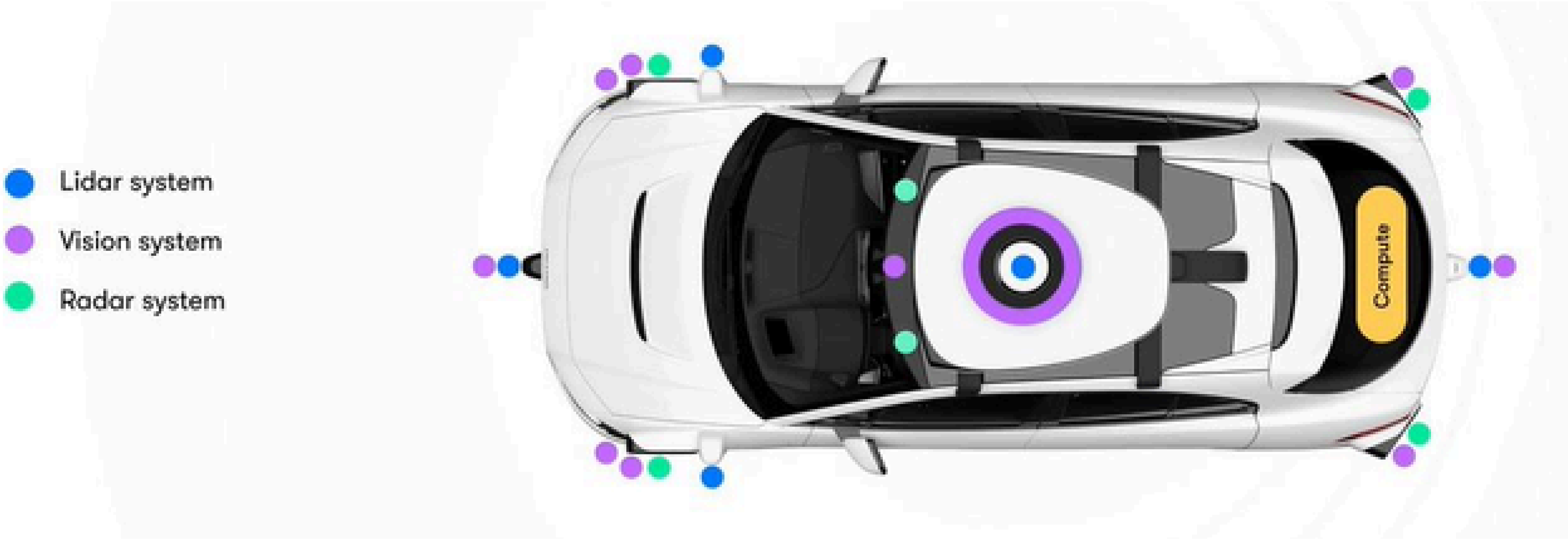

1. Must obtain special manufacturer plates issued by the Michigan Secretary of State.
2. Must report proof of insurance, business presence in Michigan, and details about the autonomous system to the Secretary of State. [21, 22]

# MVP

Problem statement → Background information → Regulations → MVP → Self-Certifications → Key learnings

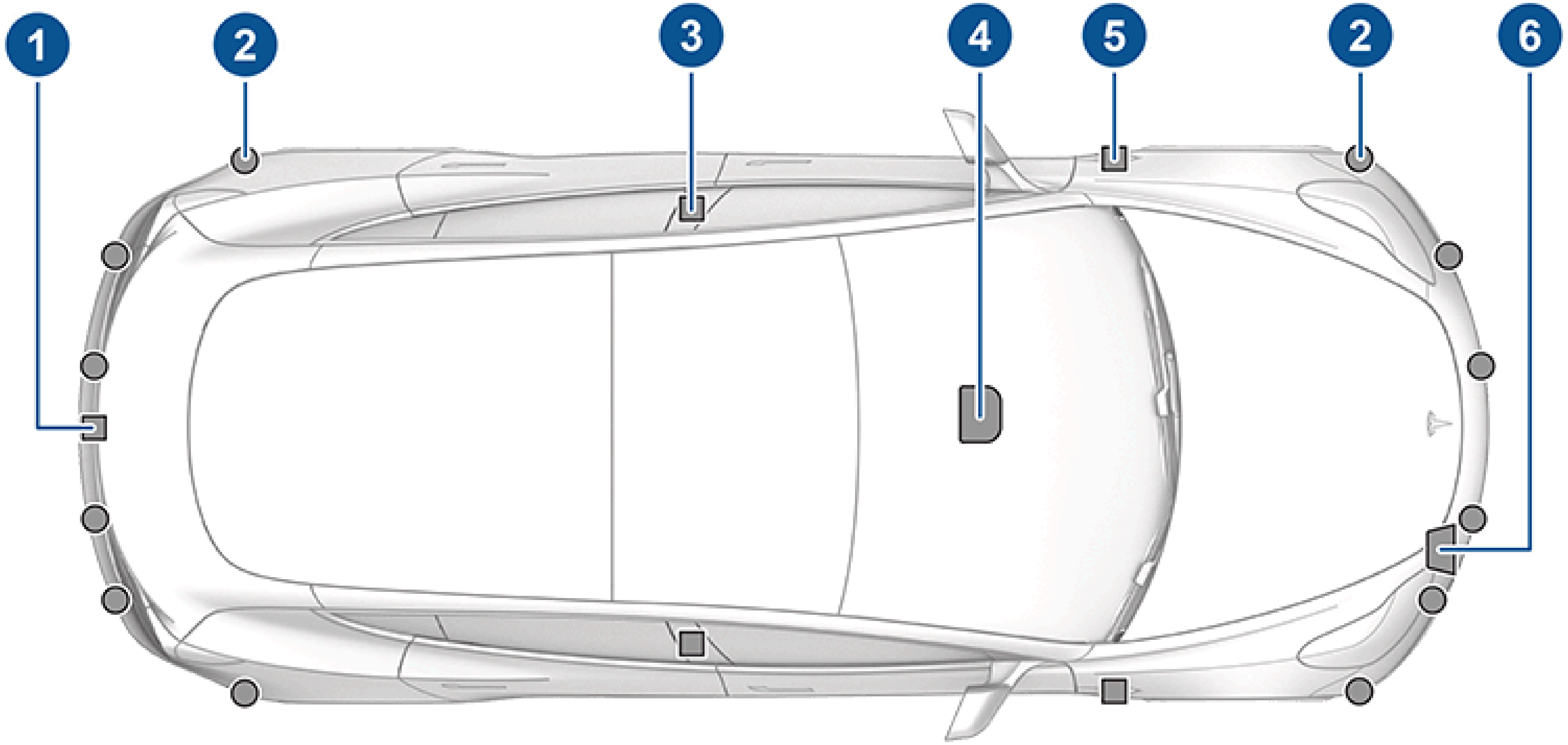
# INSPIRATION

Waymo's sensor suite:



# INSPIRATION

Tesla's sensor suite:
#6 is radar, #2 is ultrasonic



# DESIGN PRINCIPLES AND CONSTRAINTS

Camera heavy for cost and power efficiency.

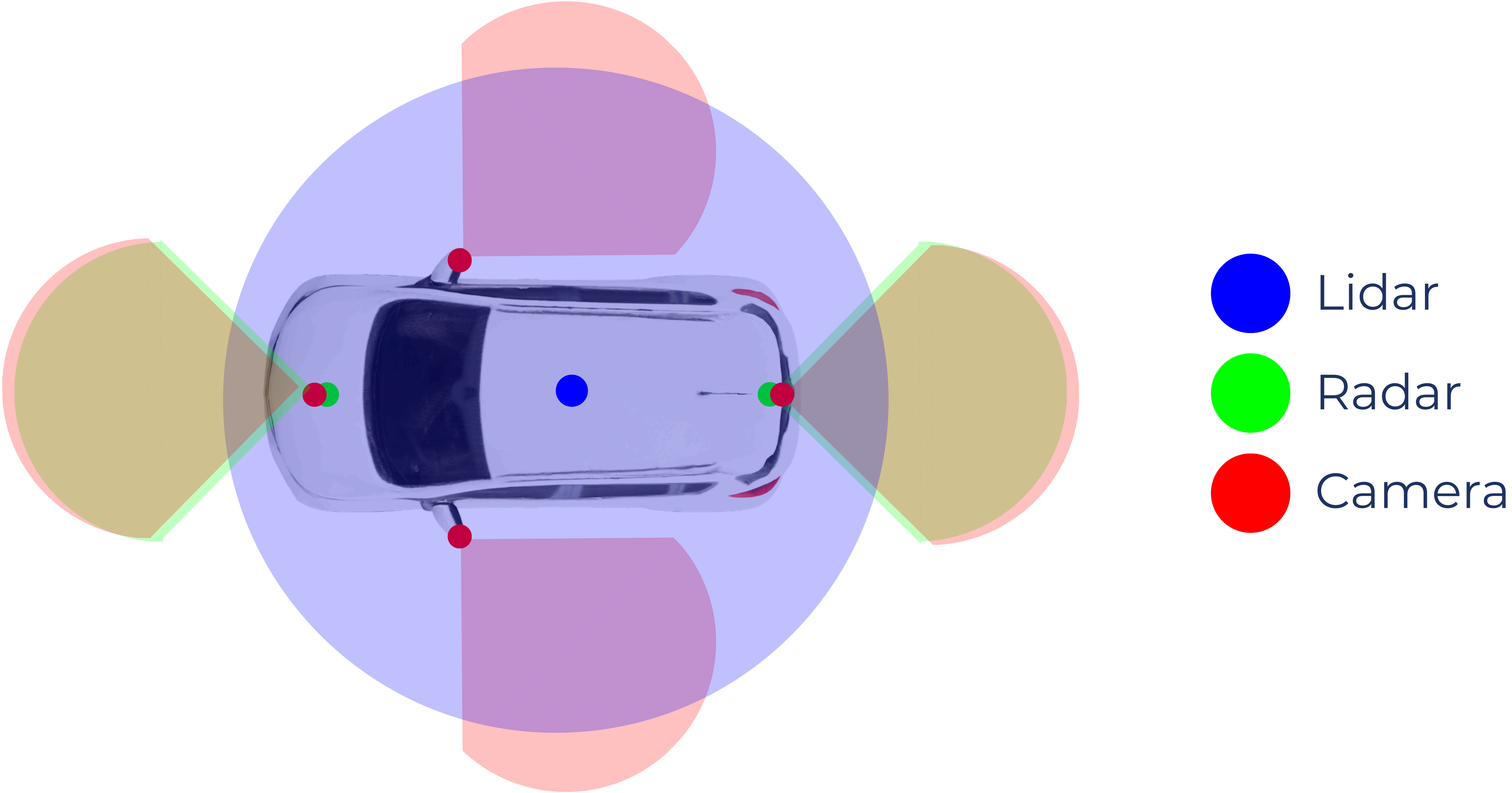
Having minimal sensor overlap, but not decreasing functionality.

4-5

Leveraging LiDAR and radar to target L4-L5 autonomy.

Compliant with all the regulations in our chosen locations.

# Our MVP



# Our MVP

Lidar

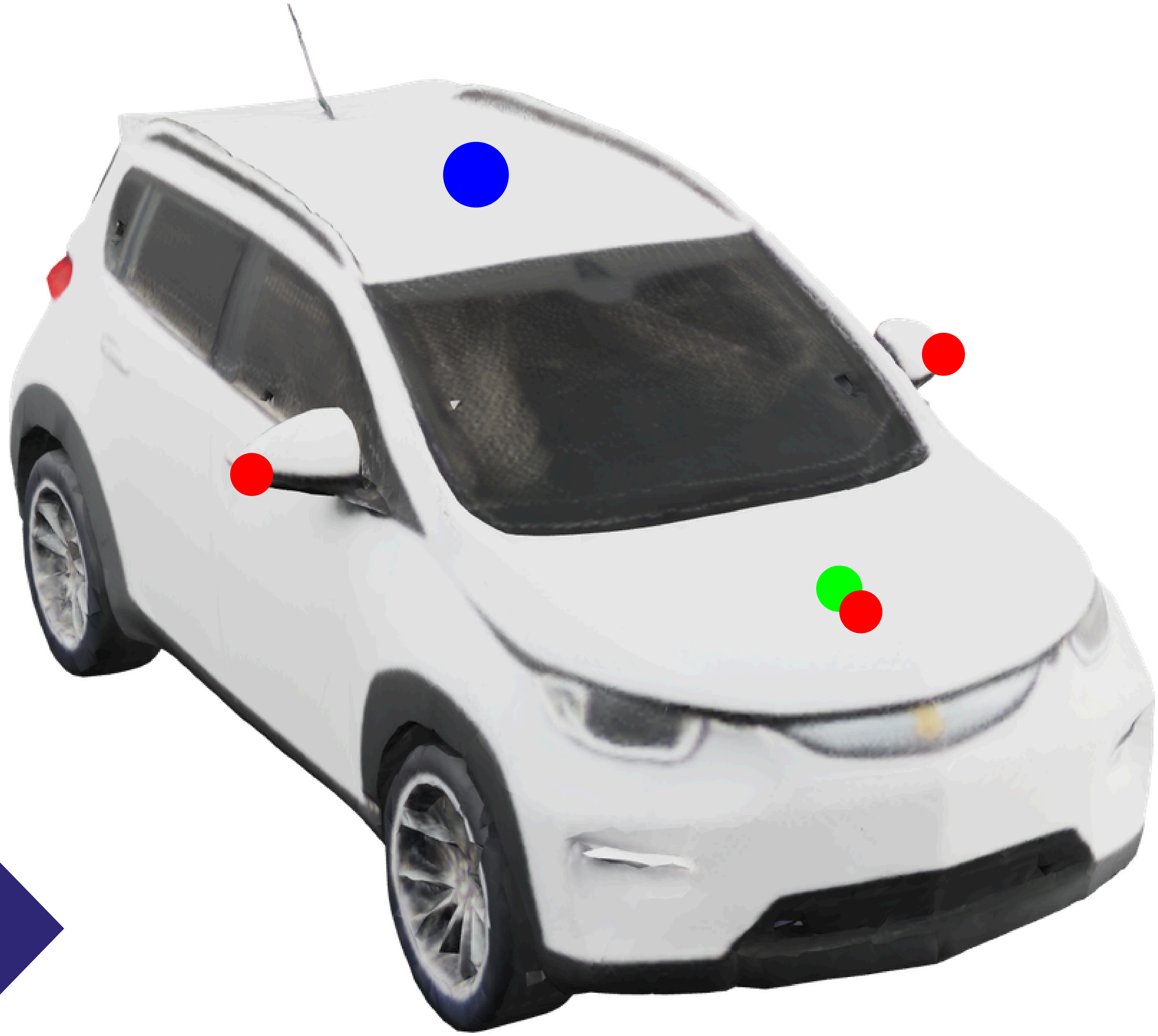
Radar

Camera

# Our MVP

Lidar
Radar
Camera

# Our MVP



**Extra Additional Features:**

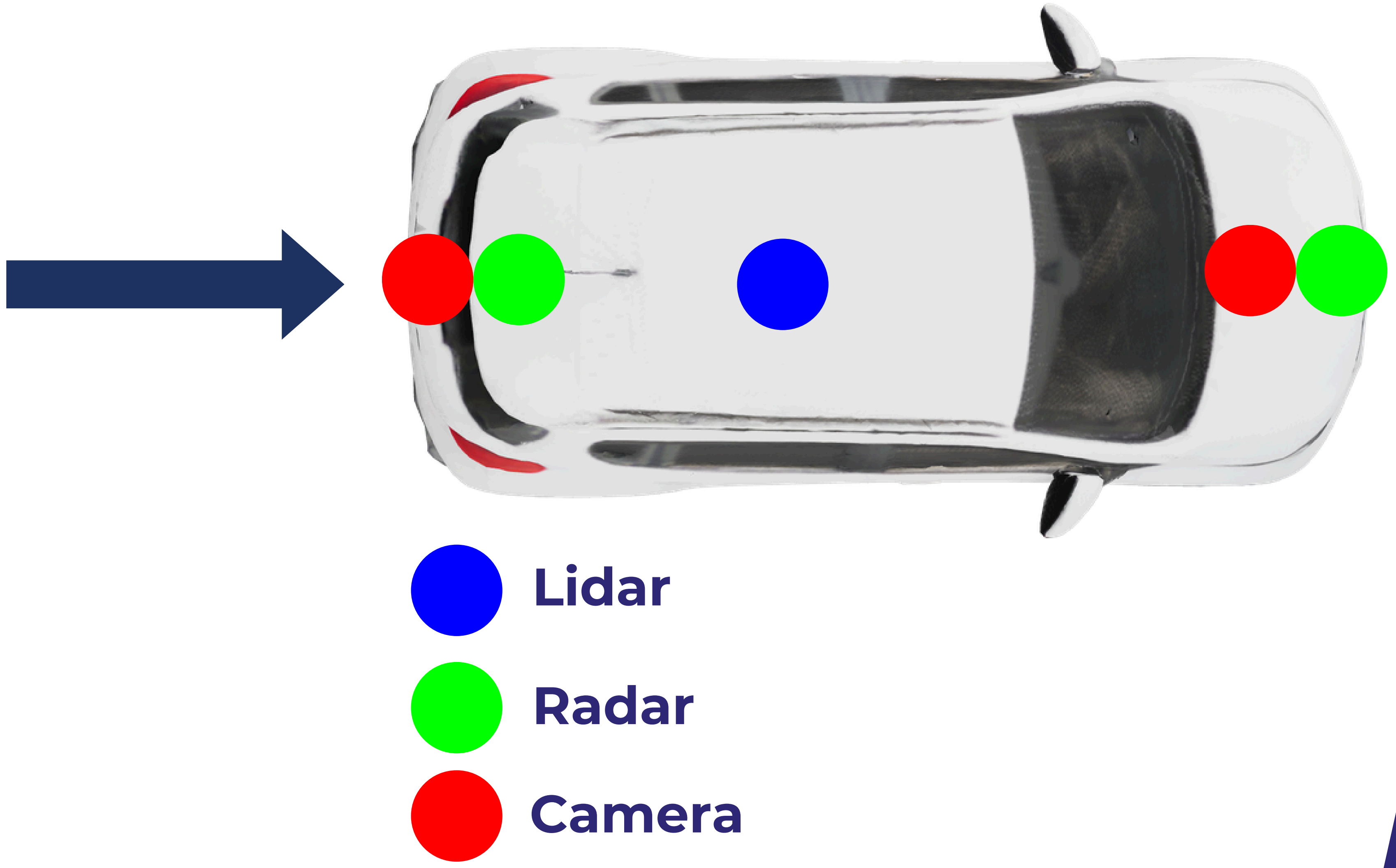
1. All the sensor data 1 minute before a crash is saved.

2. There will be an internal alerting system to indicate to the driver to take control again.

# FMVSS 111: REAR VISIBILITY

FMVSS 111: Regulation for Rear Visibility Requirements, including equipment of rearview camera that provides a clear field of view behind the vehicle when in reverse.

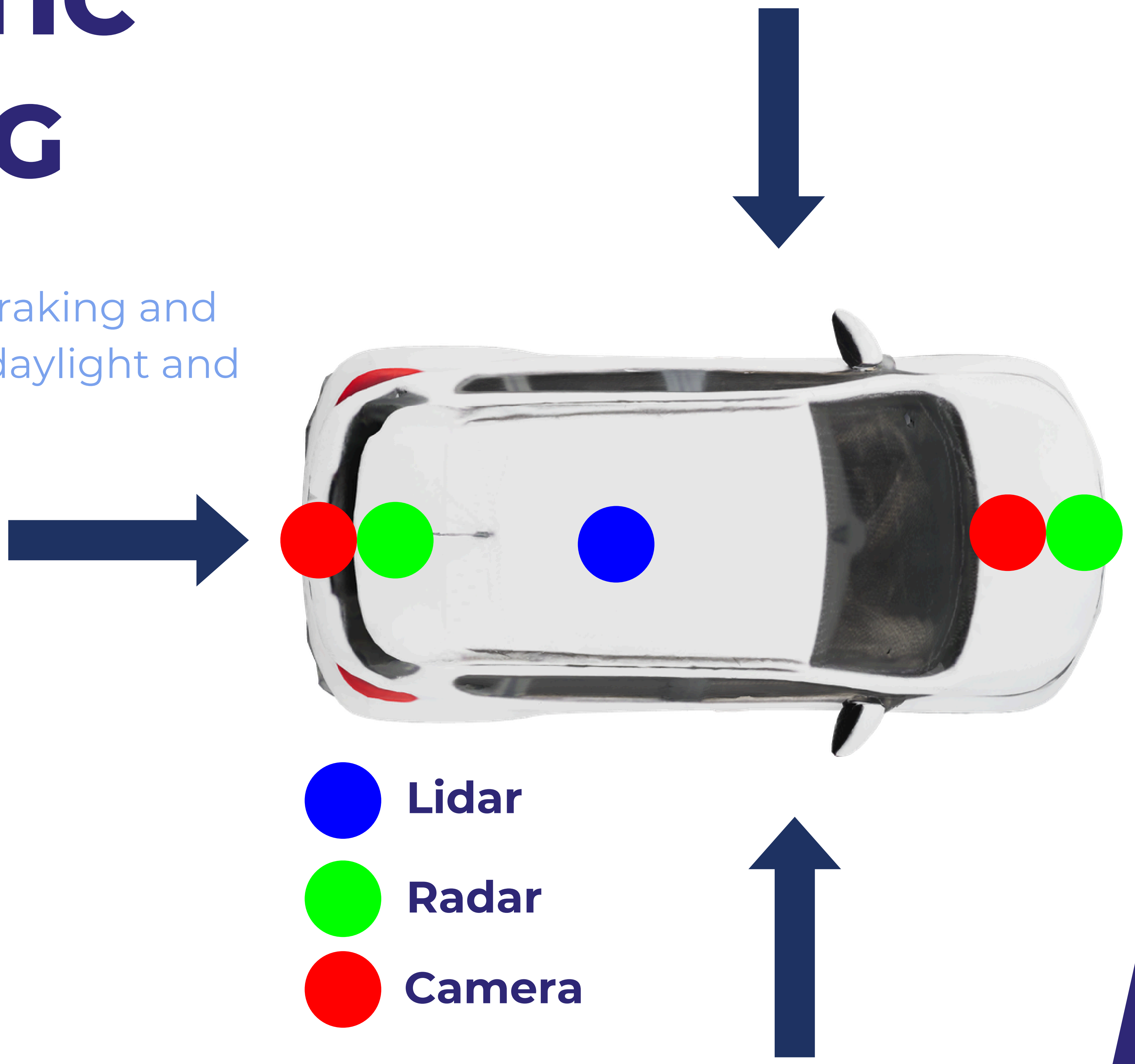
**Our sensor suite has 3 modalities of sensors for rear visibility which is more than enough to comply with this regulation, as the minimum requirement is just a clear field of view.**



# FMVSS 127: AUTOMATIC EMERGENCY BRAKING

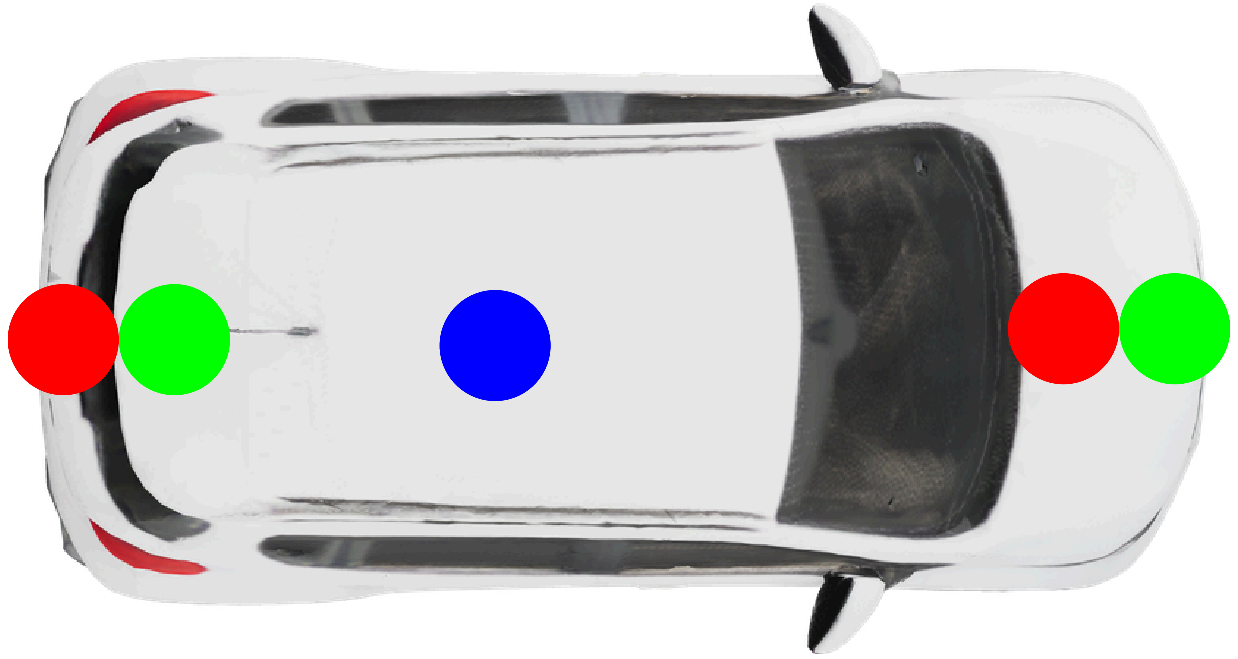
> **FMVSS 127**: Regulation for Automatic Emergency Braking and Pedestrian Automatic Emergency Braking in both daylight and darker conditions at night.

**Radar sensors are best for detection during darker/misty/low visibility scenarios, and we have 75 degrees facing the front and back.**



# FMVSS 208: OCCUPANT CRASH PROTECTION

**FMVSS 208:** Regulation for Occupant Crash Protection design and performance of safety systems in motor vehicles. Includes seat belts, airbags, crash testing, and child restraint systems.



**Our sensor suite has nothing to do with this regulation, as its regarding the functionality of the car itself. By our assumption, we will comply with this regulation.**

# KEY SPECS

Total Energy Consumption: **72W**

Total Cost of Sensors: **$2154**

Old Mileage: **259 miles**

New Mileage: **231 Miles**

# SENSOR DESIGN CHOICES

## Cost and Power Consumption

**LiDAR:** ~ **$2,000***, **~17 W**

**2x Radar:** ~ **$35****, **22.5 W**

**4x Camera:** ~ **$21****, **2.5 W**

[* Lidar price estimated from a listed device on sale]

[** Taken from R2Y4 Triple Zero]

# Our Sensors

## Valeo cameras 4x

| Brand | H x V FOV (degrees) | Price | Resolution (MP) | Power (W) |
|---|---|---|---|---|
| Valeo | 195 x 155 | 21 | 1 | <2.5 |

## Hella radars 2x

| Brand | Model | Range (m) | H x V FOV (degrees) | Price | Power (W) |
|---|---|---|---|---|---|
| Hella | SRR5G3 | 90 | 75 x 10 | 35 | 22.5 |

## Ouster OS1 LiDAR

| Brand | Model | Range (m) | H x V RES (degrees) | H x V FOV (degrees) | Price | Power (W) |
|---|---|---|---|---|---|---|
| Ouster | OS1 | 200 | 0.18 x 0.7 | 360 x 42.4 | ~2000 | 14-20 |

# Comparing our MVP

| Vehicle | Sensor Suite Power (W) | Suite kWh/mile (est.) | Vehicle kWh/mile (base) | Combined kWh/mile (approx.) |
|---|---|---|---|---|
| Tesla Model 3 | 250–300 | 0.005 | 0.24 | 0.245 |
| Waymo (Pacifica PHEV) | 2,500–3,000 | 0.08 | 0.33 | 0.41 |
| Bolt EUV + 72 W Sensor (30 mph) | 72 | 0.0024 | 0.28 | 0.2824 |
| Bolt EUV + 72 W Sensor (60 mph) | 72 | 0.0012 | 0.28 | 0.2812 |

# Self-Certification Plan & ODD Sensor Compliance

Problem statement → Background information → Regulations → MVP → Self-Certifications → Key learnings

# ODD HIGH LEVEL OVERVIEW

**Environmental**

Weather
Geolocation
Object Avoidance

**Public roads & Traffic information**

Parking
Road Types
Congestion
Road Grade

**Autonomy**

Path Planning & Execution
Lateral Controls
Longitudinal Controls

# ENVIRONMENTAL

ODD and Self Certification

### Weather

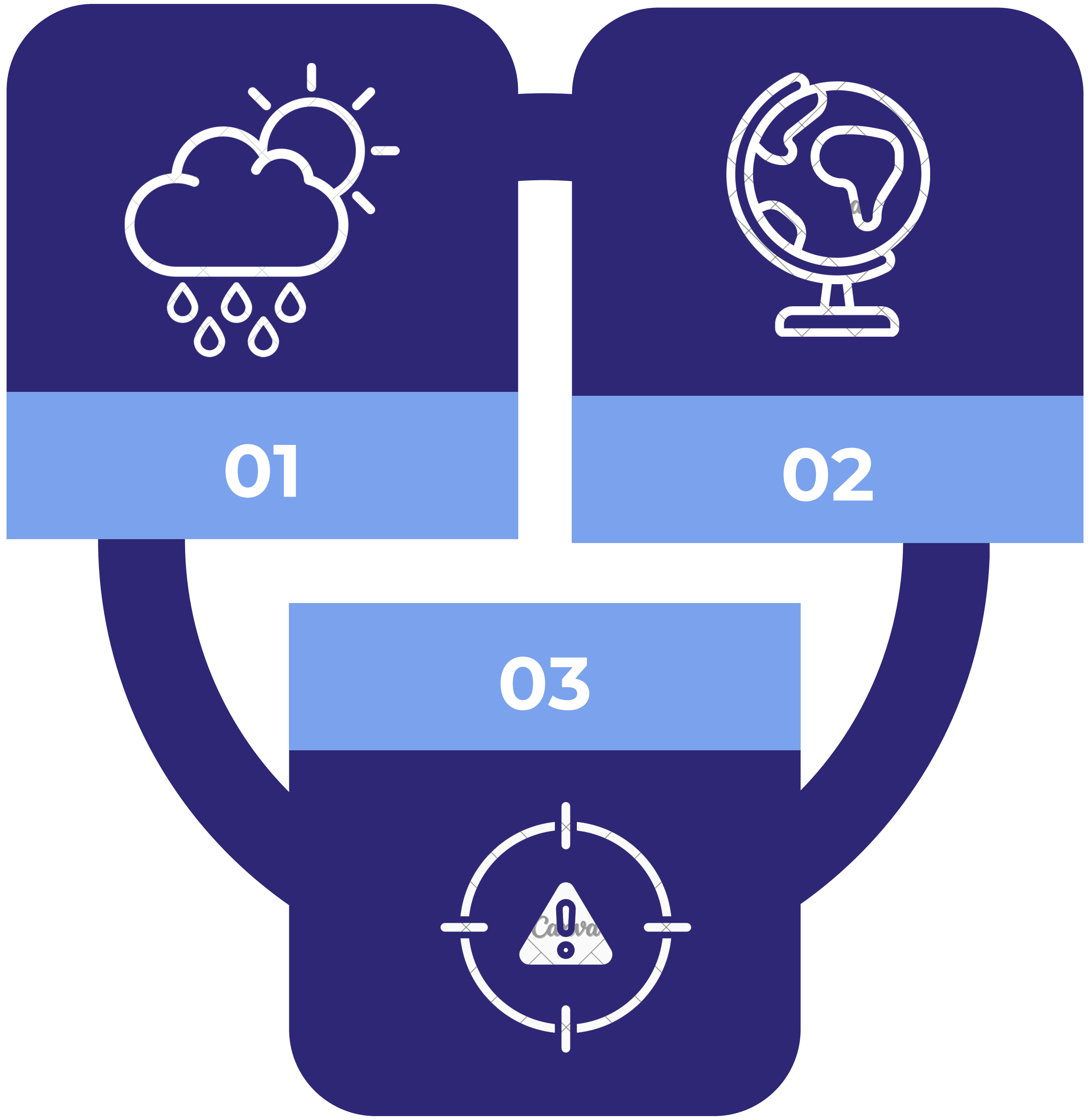
Conduct controlled track tests (10–30 trials/condition),

### Geolocation

Conduct tests across different map types (interstate/arterial/urban)

### Object Avoidance

Place standardized obstacles and run trials at multiple speeds (20–65 km/h) and lighting conditions



# PUBLIC ROADS, TRAFFIC INFORMATION

ODD and Self Certification

### Road Types

Run simulation, then a multi-turn track test for straight, curve, and roundabout segments.

### Road Grade

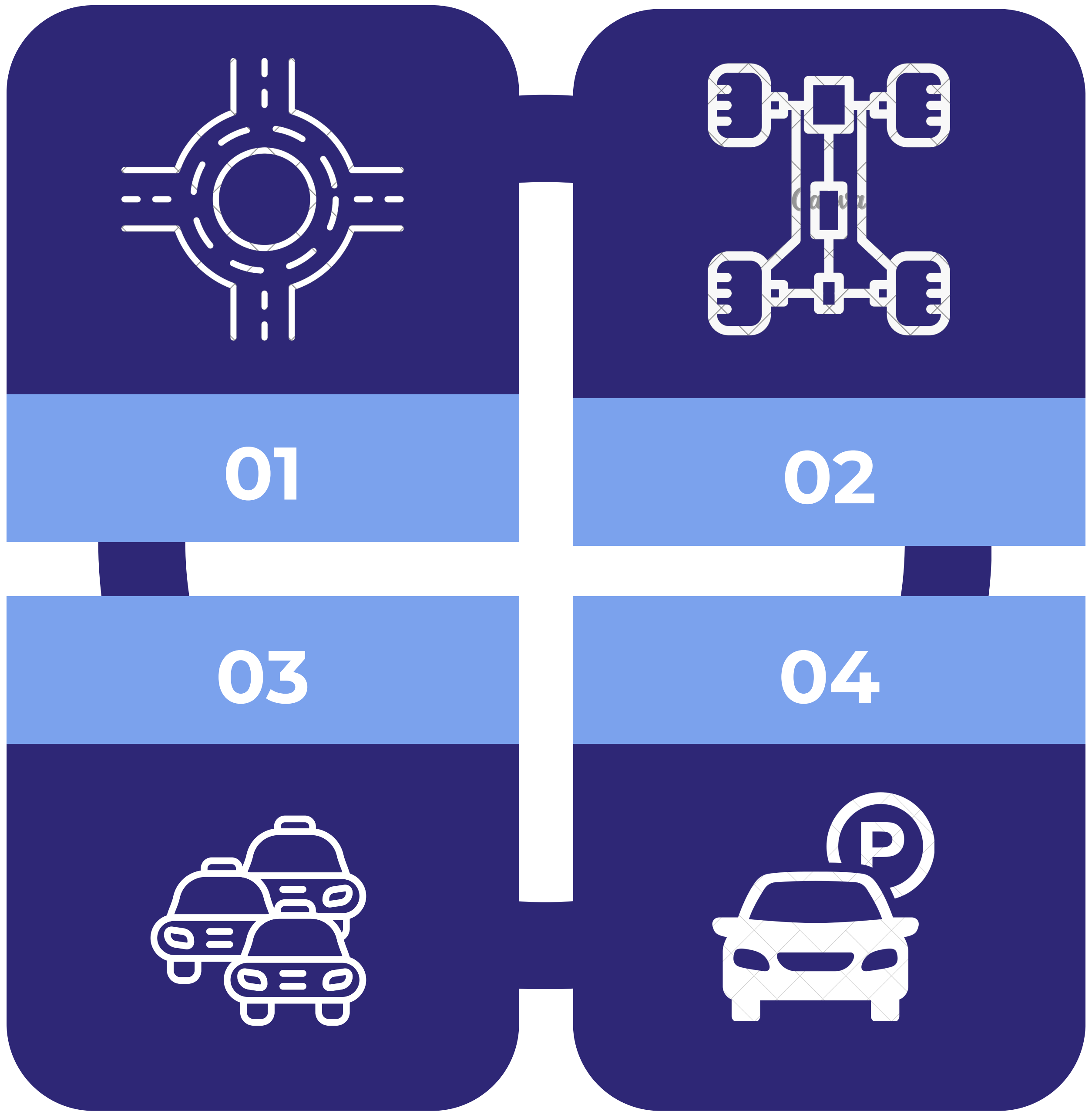
Set up chassis dyanometer device to simulate ascents/ descents at fixed speeds.

### Congestion

Run simulation, then create a test suite to simulating light to heavy congestion with principal other vehicles

### Parking

Create standard stalls for all parking categories, perform simulation tests then on track for forward & reverse parking



# AUTONOMY

ODD and Self Certification

### Longitudinal Controls

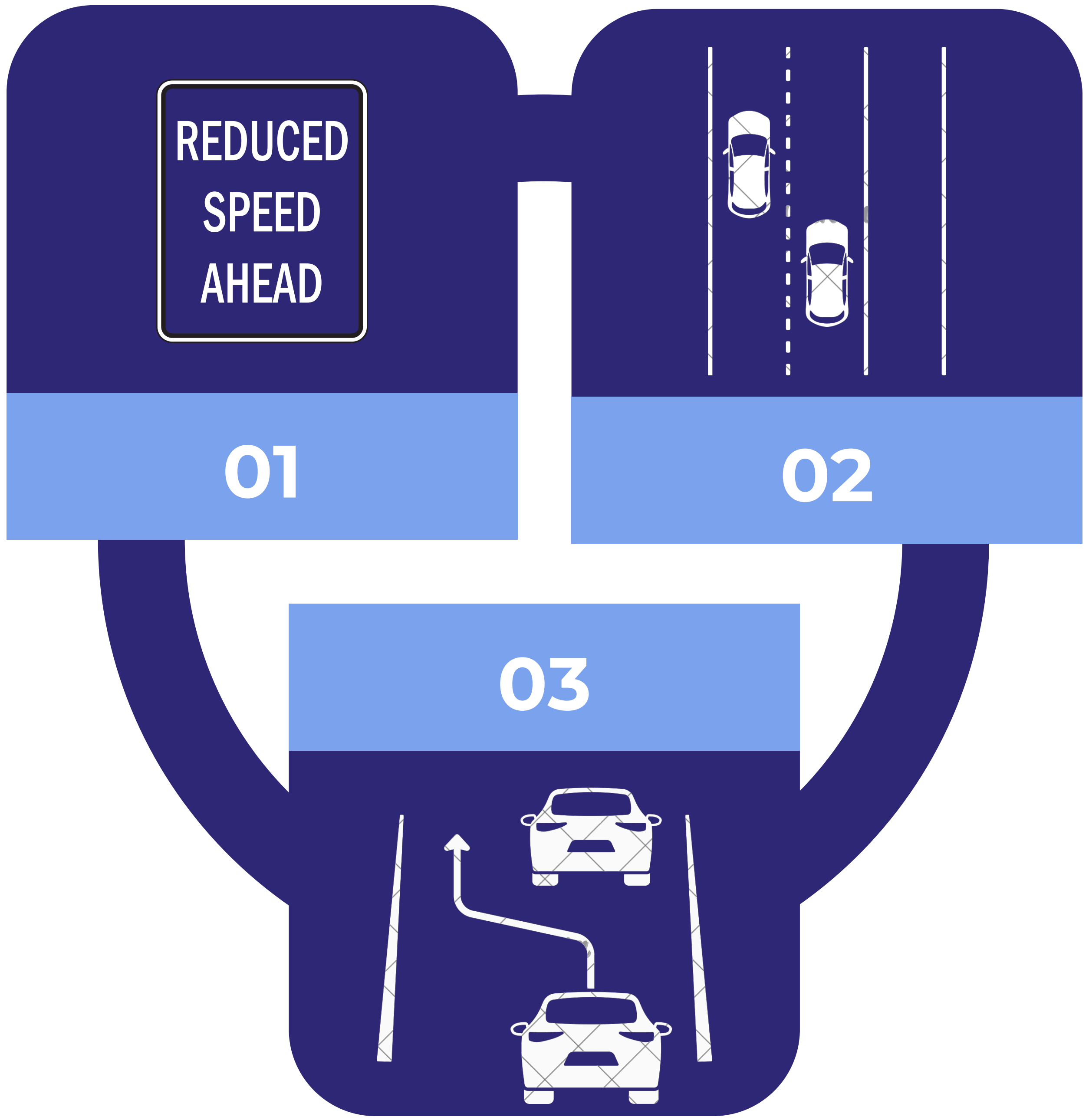
Verify in simulation that the vehicle can safely follow, detect signs, and adjust speed before track testing.

### Lateral Controls

Verify in simulation vehicle's lane centering, lane changes, and turn capabilities in simulation before track testing

### Path Planning & Execution

Run simulation, then a multi-turn track test with lane changes, merges, detour, random signals, and emergency-vehicle events



# SELF-CERTIFICATION PLAN

## We validate system compliance with:

FMVSS 111 FMVSS 127 FMVSS 208

## We validate operational compliance with the following ODDs:

| Environmental | Public roads & Traffic information | Autonomy |
| --- | --- | --- |
| **Weather**<br>**Geolocation**<br>**Object Avoidance** | **Parking**<br>**Road Types**<br>**Congestion**<br>**Road Grade** | **Path Planning & Execution**<br>**Lateral Controls**<br>**Longitudinal Controls** |

## This certification is supported by:

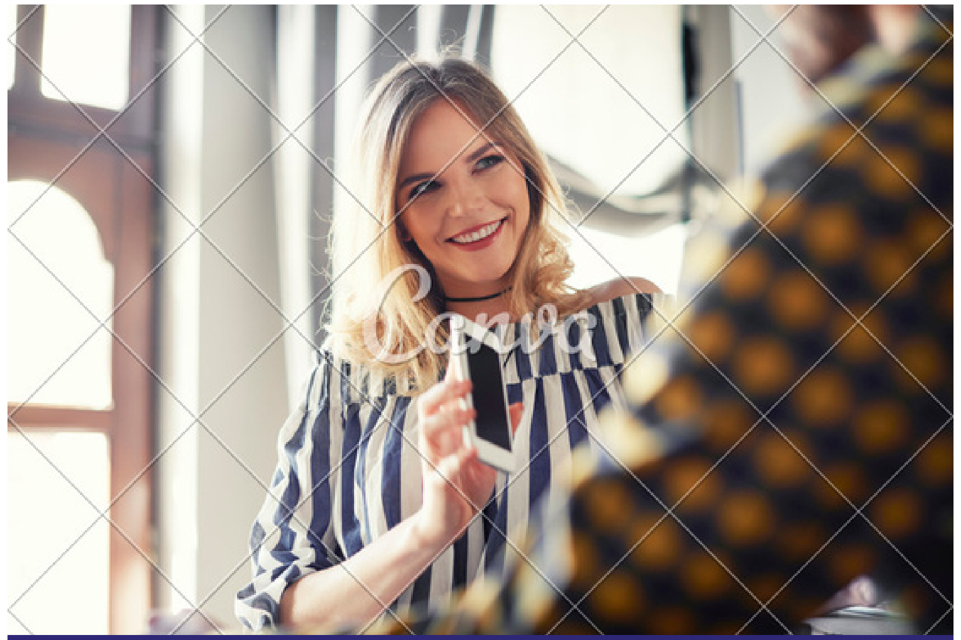
- Verified ODD-level test evidence
- Sensor coverage and performance validation
- Simulation, track, and real-world testing

# CHALLENGES & KEY LEARNINGS

Presentations are tools that can be used as lectures, speeches, reports, and more. It is mostly presented before an audience.

### 01.Advantage

It is mostly presented before an audience.



### 02.Advantage

It is mostly presented before an audience.

### 03.Advantage

It is mostly presented before an audience.

# THE TEAM

THANK YOU

**Jamie Chastain**
Manager

**Isabel Mercado**
Manager

**Shawn Garcia**
Manager

**Yanis Petros**
Manager

# TITLE

Test Execution → Data Analysis
→ Compliance Documentation

**Competitor 01**

It is mostly presented before an audience.

**Competitor 02**

It is mostly presented before an audience.

**Competitor 03**

It is mostly presented before an audience.

**Competitor 04**

It is mostly presented before an audience.

# MARKET SIZE

Presentations are tools that can be used as lectures, speeches, reports, and more. It is mostly presented before an audience.

**$500 M**

### Total Available Market

It is mostly presented before an audience.

**$100 M**

### Serviceable Available Market

It is mostly presented before an audience.

**$50 M**

### Serviceable Obtainable Market

It is mostly presented before an audience.